# Tanay Mevada

*Data Engineer*

tanaymevada174@gmail.com | 551-323-3630 | Jersey City, NJ | linkedin.com/in/tanay-mevada-05301a145

## Professional Summary:

Data Engineer with over 4 years of experience building scalable cloud pipelines, Document AI workflows, and ML-powered processing systems. Skilled in GCP, AWS, Python, LangChain, PGVector, BigQuery, and ETL/ELT development. Proven track record implementing OCR-based extraction, semantic search, and RAG pipelines for faster retrieval, improved accuracy, and reduced manual effort. Strong in Spark, Kafka, Databricks, and data warehousing with Snowflake, PostgreSQL & NoSQL. Adept at turning raw data into high-value insights using NLP, embeddings, and Generative AI to drive business efficiency and intelligent automation.

## Technical Skills:

- **Programming & Scripting:** Python, SQL, R, Shell/Bash
- **Cloud Platforms & Services:** AWS (Lambda, Glue, S3, EC2, DynamoDB, API Gateway), Google Cloud Platform (BigQuery, Cloud Functions)
- **Big Data & Distributed Systems:** Apache Spark, Kafka, Hadoop/Hive, Databricks, Delta Lake, Lakehouse Architecture
- **Databases & Storage:** PostgreSQL, MySQL, Snowflake, BigQuery, NoSQL (DynamoDB, MongoDB), Data Warehousing
- **AI / ML / GenAI:** TensorFlow, PyTorch, Keras, Scikit-Learn, LangChain, LlamaIndex, NLP, NER, Semantic Chunking, RAG Pipelines, Generative AI (LLMs), OpenAI API, OCR, Entity Extraction, Supervised Learning and Classification Models
- **Data Engineering & ETL Pipelines:** Airflow, DBT, ETL/ELT Pipeline Development, Data Modeling, Batch & Streaming Pipelines, Data Quality & Validation, Change Data Capture (CDC)
- **Vector Search & Retrieval:** PGVector, Embedding Search, Document Classification, Document Similarity, Hybrid Retrieval Systems
- **Analytics, BI & Visualization:** Power BI, Looker, Tableau, QuickSight, Matplotlib, Dashboarding and Data Storytelling
- **Containerization, CI/CD & DevOps:** Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, CI/CD Pipelines
- **Testing & Automation:** Pytest, Unit Testing, API Testing, Automated Data Pipeline Testing
- **Version Control & OS:** Git, Linux, Windows
- **IDEs & Development Environment:** VS Code, PyCharm, Jupyter Notebook

## Professional Experience:

**Data Engineer** — July 2023 – Present
**Etsy, NY**

- Designed and built scalable Document AI pipelines using Google Cloud Vertex AI and BigQuery, improving OCR accuracy, data extraction efficiency, and document classification speed.
- Developed semantic chunking and entity extraction algorithms on Vertex AI, enhancing Named Entity Recognition (NER) and document processing accuracy.
- Streamlined testing workflows by integrating Pytest with Google Cloud Functions, reducing manual efforts and increasing overall testing efficiency by 40%.
- Created an AI-powered document search system leveraging PG Vector, LangChain, and GCP AI tools, enabling faster and more precise document retrieval.
- Optimized data retrieval with Retrieval-Augmented Generation (RAG) pipelines in BigQuery, improving performance for document comparison and content generation.
- Developed and deployed APIs using Google Cloud API Gateway, ensuring seamless document classification and content retrieval with automated validation.

**Data Engineer** — June 2020 – July 2022
**PetPooja**

- Designed and implemented scalable data pipelines utilizing AWS S3, Lambda, and Glue, improving data processing and storage efficiency for large-scale operations.
- Developed event-driven real-time data integration systems using Lambda and API Gateway, reducing processing delays and enhancing third-party integrations.
- Integrated DynamoDB with various AWS services, ensuring seamless data flow, system reliability, and optimized database performance.
- Built and deployed Python-based microservices using Django, enabling efficient data handling and smooth integration with front end applications.
- Automated structured data extraction from forms using AWS Textract, improving accuracy and minimizing manual intervention.
- Designed real-time BI dashboards with QuickSight, providing data-driven insights and actionable reports for key decision-making

## EDUCATION

**Master of Science in Data Science**
**Saint Peter's University**
Relevant Coursework: Data Analysis, Data Mining, Big Data & Data Management, Data Visualization

**Bachelor of Engineering in Information Technology**
**Gujarat Technological University**
Relevant Coursework: Database Management Systems, Object-Oriented Programming

## Projects

**Expense Tracker | Python, Django, Web Application**
- Developed a full-stack expense tracking web app using Django to streamline personal financial management
- Implemented modules for expense logging, category-wise analysis, and spending trend visualization, improving budgeting efficiency and user insights.

**Stock Price Prediction Using Weather Data | Python, Time Series Analysis (Random Forest, Linear Regression, ARIMA)**
- Built predictive models to forecast Dow Jones Industrial Average (DJIA) trends by integrating historical stock and NYC weather data
- Conducted time-series modeling and feature correlation to improve prediction accuracy and highlight weather-driven volatility.

**Yelp Review Analysis | Python, SVM, Jupyter Notebook**
- Applied Support Vector Machine (SVM) models to analyze NYC restaurant reviews, identifying sentiment patterns and customer satisfaction drivers
- Delivered actionable insights to enhance business performance and improve online reputation management.

**Anime Recommender System | Python, KNN, Jupyter Notebook**
- Designed a content-based recommender system using K-Nearest Neighbors (KNN) to suggest Anime titles based on user viewing history
- Improved recommendation precision and user engagement through data preprocessing and similarity scoring.